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**Portfolio · Helix family · vasic-digital · Server
Factory**

A unified, evidence-based portfolio — the Helix product family, the vasic-digital utility fleet, and the Server Factory toolchain, on a shared engineering Constitution.

140+

REPOSITORY FLEET

43

LLM PROVIDERS

25LINUX DISTROS
PROVISIONED**439**PASSING TESTS · MAIL
SERVER FACTORY

1. Overview — the whole picture first

This is a single, unified portfolio used by both vasic.digital and milosvasic.ru. It is organized as a fleet: **large product applications** on top of **dozens of small, decoupled, independently-tested modules**, all governed by a shared engineering **Constitution** and verified by an **anti-bluff QA** discipline. The dominant language is **Go**, with Kotlin/KMP, TypeScript/React, Python, Swift, and Shell used where they fit best.

Everything below is presented in priority order:

1. **Governance & QA pillars** — HelixConstitution, HelixQA (the discipline that makes the rest trustworthy).
2. **The Helix product family** — the AI-development lifecycle (HelixTrack first).
3. **LLM infrastructure** — provider abstraction, orchestration, verification.
4. **vasic-digital utilities** — product-grade standalone tools.
5. **Server Factory** — infrastructure-automation heritage (ranked last).

The unifying thesis: **a feature is done only when a real user can use it and there is captured evidence to prove it.**

2. Governance & QA pillars

- **HelixConstitution** — a universal, project-agnostic engineering rulebook shipped as a Git submodule and inherited across a 140+-repository fleet. Anti-bluff evidence gates, false-positive immunity, data/host safety, coverage discipline; extend-but-never-weaken inheritance; propagation gates that grep for required clauses fleet-wide; every gate paired with a mutation test proving it isn't a sham.
- **HelixQA** — anti-bluff QA orchestration (Go). Written YAML test banks plus fully-autonomous LLM-and-computer-vision QA sessions across Android, Android TV, Web, and Desktop; no PASS without captured evidence (screenshots, logcat, video, stack traces). The Constitution's mandated QA test-type (§11.4.169).

3. The Helix product family

- **HelixTrack** — a JIRA alternative for the free world; flagship of the Helix-Track line.
- **HelixAgent** — ensemble LLM service: multiple models debate and ship the answer they agree on, with verification-based provider selection.
- **HelixCode** — enterprise-grade distributed AI development platform; divides work across SSH-managed workers with automatic checkpoint/rollback; REST, CLI, TUI, MCP interfaces.

- **HelixCluster** — a distributed operating system for AI compute, from datacenter GPUs to edge handhelds, under one control plane.
- **HelixBuilder** — an AI-powered pipeline for building applications, one category at a time.
- **HelixSkills** — a governed, constitution-backed skills system for CLI AI agents (skills, MCP tool servers, Claude Code plugins).
- **HelixSpecifier** — spec-driven development that scales its ceremony to the work.
- **HelixMemory** — one memory brain for AI agents, fusing four best-in-class engines.
- **HelixTranslate** — verified-model book translation; honest by design, never a silent fallback.
- **HelixTerminator** — a zero-trust terminal platform: every SSH session secured, shared, and AI-assisted.
- **HelixGitpx** — federated Git across a dozen hosts; one source of truth, mirrored everywhere.
- **HelixOTA** — universal, decoupled over-the-air updates; zero-brick by design.
- **HelixPlay** — turn any GPU machine into your own cloud-gaming appliance.
- **Helix-Flow** — Helix-platform inference product. *UNVERIFIED / blocked on source: its public repo currently has only a one-line README; presented at product depth only when real documentation exists.*

4. LLM infrastructure

- **HelixLLM** — one binary, six modes: OpenAI- and Anthropic-compatible inference over HTTP/3, local llama.cpp, scored fallback chain, RAG pipeline, ReAct agents.
- **LLMProvider** — one interface, 43 providers, with circuit breakers, retries, and health built in.
- **LLMOrchestrator** — one control plane for every headless CLI coding agent (OpenCode, Claude Code, Gemini, Junie, Qwen Code).
- **LLMsVerifier** — verify, monitor, optimize: the single source of truth for LLM/provider/verification metadata, with a mandatory model comprehension gate.

5. vasic-digital utilities

- **Catalogizer** — multi-protocol (SMB/FTP/NFS/WebDAV/local), encrypted, self-hostable media collection management; Go/Gin API + React UI; resilient offline-tolerant monitoring; built on 21 `digital.vasic.*` submodules.
- **Courses-Creator** — markdown-to-video course pipeline; multi-LLM enrichment, TTS (Bark/SpeechT5), desktop/mobile/web players; graceful no-API-key mode.
- **VisionEngine** — decoupled Go toolkit fusing classic CV with multi-provider LLM vision; navigation graphs with BFS + DOT/JSON/Mermaid export; OpenCV build-tag-gated.
- **DocProcessor** — documentation-to-feature-map with verification-coverage tracking; LLM or heuristic/offline extraction; Apache-2.0.
- **Docs Chain** — content-hashed, bidirectional, atomic doc/DB sync (Salsa-style incremental recompute over a DAG). *Phases 1–5 GREEN; Phases 6–7 PLANNED.*

- **Herald** — reliable multi-channel notifications with natural-language, three-tier intent resolution (command → LLM → clarify); first Docs Chain consumer.
- **task_bridge** — decoupled, bidirectional task/board sync (SQLite SSoT [?] docs [?] ClickUp). *P1 scaffold — sync logic not yet implemented.*
- **Vasic Digital Reusable Module Suite** — the `digital.vasic.*` “standard library”: infrastructure primitives, AI building blocks, and defensive-LLM guardrails, plus a Kotlin Multiplatform mirror. *Several org repos are SCAFFOLD/WIP — flagged, not shipped.*

6. Server Factory (infrastructure-automation heritage — ranked last)

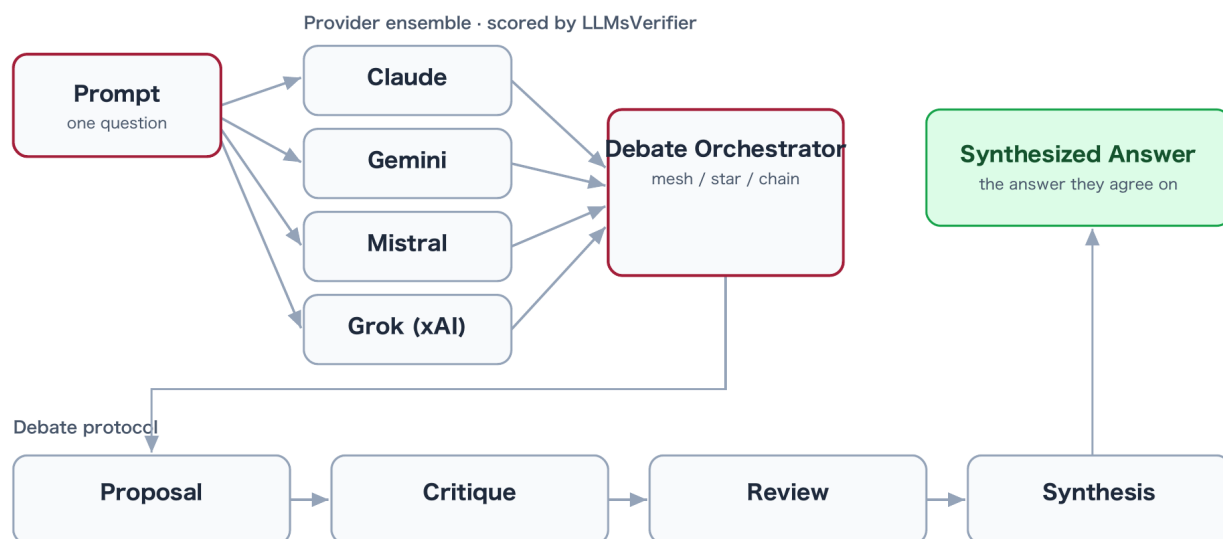
- **Mail Server Factory** — declarative JSON → fully-provisioned, Dockerized mail server across 12 connection types and 25 Linux distributions; enterprise security; reports 439 passing tests and a clean SonarQube gate. Flagship of the Server-Factory org.
- **Server Factory Core Framework** — the shared Kotlin engine every factory builds on.
- **Qemu-Utils** — QEMU VM images managed like artifacts: download/cache/run, compress/publish, bridge/TAP networking, ISO installs; Linux + macOS.
- **Parallels-Utils** — Parallels (macOS) VM image compression, publishing, and retrieval via simple settings files.
- **Server Factory — Additional Components** — service factories (Web/SonarQube/Caching-Proxy), Definitions packs, and Utils. *Service factories are placeholder-documented — UNVERIFIED / early-stage.*

7. Technology index (evidence-based)

- **Languages:** Go (dominant), Kotlin & Kotlin Multiplatform, TypeScript, Python, Swift, Shell; PL/pgSQL; TLA+ (formal specs in `helix_cluster`).
- **AI / LLM:** 40+ provider access, MCP, RAG, vector DBs & embeddings, planning (HiPlan/MCTS/Tree-of-Thoughts), LLMops, benchmarking (SWE-bench/HumanEval/MMLU), TTS (Bark/SpeechT5), computer-vision + LLM-vision, guardrails/red-team.
- **Backend:** Gin, gRPC + Protobuf, HTTP/3 (QUIC), WebSockets, Angular, React, Kafka/RabbitMQ.
- **Data:** PostgreSQL, SQLite, SQLCipher, Redis, Neo4j, ClickHouse, MinIO/S3/GCS/Azure.
- **Infra / DevOps:** Docker & Compose, Kubernetes + Helm, Prometheus + Grafana, OpenTelemetry, QEMU/Libvirt/Parallels; GitHub Actions, Gradle, Make.
- **Testing / QA:** HelixQA, Challenge harnesses with mutation gates, `go test -race`, visual-regression tooling, ADB device testing, SonarQube, security scanning (semgrep/gosec/trivy/snyk/gitleaks/nancy), TLA+ model checking.

Selected architecture

HelixAgent — ensemble debate flow



HelixAgent — an ensemble LLM service where providers debate under a four-stage protocol and ship the answer they agree on, scored by LLMsVerifier.

HelixCluster — seven-layer stack

14 control-plane
microservices

L7 · Federation & Observability

L6 · Security & Attestation (SPIFFE, PQ-E2EE)

L5 · Sessions & Interactive Terminal

L4 · AI Inference Routing

L3 · Omega Scheduler (two-level)

L2 · Consensus & Membership (Raft + SWIM)

L1 · Node Runtime & Transport (gRPC, WireGuard)

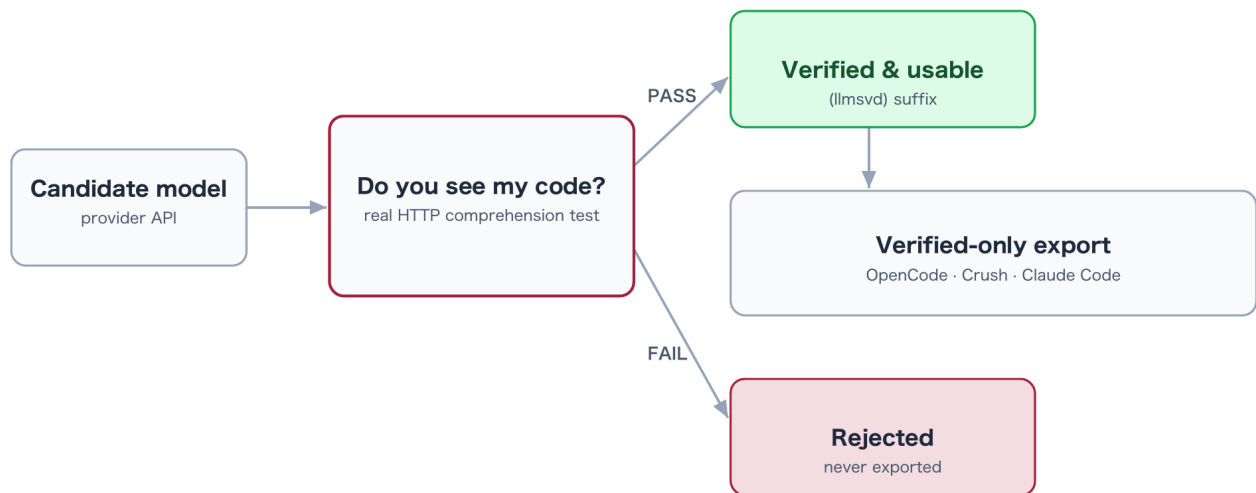
heterogeneous
nodes T1–T8

L0 · Hardware Substrate (GPU → edge SBC)

Node tiers T1 (datacenter GPU) → T8 (handheld), unified under one control plane

HelixCluster — a distributed operating system for AI compute, from datacenter GPUs to edge handhelds, under one control plane.

LLMsVerifier — mandatory verification gate



LLMsVerifier — verify, monitor, optimize: the single source of truth for LLM/provider/verification metadata, gated by a mandatory model-comprehension check.